



Calculating theoretical probabilities from two-way tables (two events)

Task A: Theoretical probabilities

1) This spinner is going to be spun once.

a) Fill in the outcome table for the two events: less than 12 and prime.



	less than 12	not less than 12
prime	2 5 7 11	13
not prime	4 8 10	16 20

b) Find $P(\text{less than 12 and prime})$

$$= \frac{4}{10} = \frac{2}{5} = 0.4 = 40\%$$

c) Find $P(\text{not prime and not less than 12})$

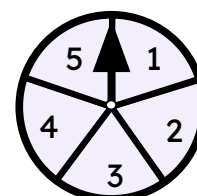
$$= \frac{2}{10} = \frac{1}{5} = 0.2 = 20\%$$

d) Find $P(\text{prime and not less than 12})$

$$= \frac{1}{10} = 0.1 = 10\%$$

2) This spinner is going to be spun twice and then the product of the two numbers is found.

a) Complete the outcome table to show the possible products.



		spin 1				
		1	2	3	4	5
spin 2	1	1	2	3	4	5
	2	2	4	6	8	10
	3	3	6	9	12	15
	4	4	8	12	16	20
	5	5	10	15	20	25

b) Find $P(\text{square and factor of 36})$

$$= \frac{5}{25} = \frac{1}{5} = 0.2 = 20\%$$

c) Find $P(\text{even and less than 18})$

$$= \frac{14}{25} = 0.56 = 56\%$$

3) This two-way table shows the frequencies of some sports balls categorised by their shape and whether they have stitching on the surface.

e.g. an american football is not spherical but has stitching, whereas a golf ball is spherical but doesn't have stitching.



	spherical	not spherical	total
stitching	4	1	5
no stitching	11	2	13
total	15	3	18

A ball is selected at random.

a) Find $P(\text{spherical and has stitching})$

$$\frac{4}{18}$$

b) Find $P(\text{spherical and does not have stitching})$

$$\frac{11}{18}$$

Task B: Probabilities for combined events

1) This outcome table shows the sample space {1, 2, 3, 4, 5, 6, 7, 8, 9}.

An outcome is going to be randomly selected.

Find the probability it is:

a) even and a factor of 18 or odd $\frac{7}{9}$

b) odd and a factor of 18 or even $\frac{7}{9}$

c) a factor of 18 or even and not a factor of 18 $\frac{7}{9}$

d) not a factor of 18 or even $\frac{6}{9}$

	factor of 18	not a factor of 18
even	2 6	4 8
odd	1 3 9	5 7

